

DevOps is the Future of Software Development

DevOps scores over legacy, monolithic and agile software development.

This article discusses the various stages of the DevOps software development cycle and delineates the appropriate FOSS tools that can be used at each of the stages.



DevOps is a method of software engineering that combines both software development (Dev) and information technology operations (Ops) to create high quality software within a short period of time. DevOps shortens the time taken for software development through continuous delivery and the integration of code. As the name suggests, DevOps is a combination of development and operations. Traditionally, software development is done by a team of stakeholders comprising business analysts, software engineers, programmers and software testers. The software development is done

by the development team through a software life cycle comprising various stages like customer requirement(s), planning, modelling, construction and deployment. However, such a development cycle could take a lot of time and collaborative effort to successfully deliver software to the customer. This is also known as agile software development.

DevOps can help overcome the drawbacks of the agile software development life cycle. In the agile method, to develop software, every stage of the life cycle needs to be completed prior to the final release and that could take a lot of time for the software to reach maturity. In the

case of DevOps, instead of delivering the whole software, small chunks of code are updated, and the updated software is released to the operations team continuously, which speeds up the software development cycle. The DevOps life cycle can be automated using various development tools and requires less manual activity. The DevOps concept first originated in 2008 during a discussion between Patrick Debois and Andrew Clay Shafer. However, the idea only started to spread in 2009 with the advent of the first DevOps Days event held in Belgium. Now, most tech giants like Facebook, Google, Amazon, Netflix, etc, have adopted the DevOps culture.

